Customer Segmentation Project Report

1. Introduction

Customer segmentation is the process of dividing customers into different groups based on their behaviour and characteristics. This helps businesses understand their customers better and create targeted strategies.

2. Objective

The main objective of this project is to segment customers based on their Annual Income and Spending Score using clustering techniques.

3. Tools and Technologies Used

- Python
- Scikit-learn (for clustering)
- Matplotlib (for visualization)

4. Methodology

In this project, the K-Means clustering algorithm is used to group customers into different segments.

Steps followed:

- Loaded the dataset
- Selected important features (Income & Spending Score)
- Applied K-Means clustering
- Assigned cluster labels to each customer
- Visualized the segments using a scatter plot

5. Results

The customers were successfully divided into 5 different clusters based on their income and spending behaviour.

6. Analysis

- Customers with high income and high spending are premium customers
- Customers with low income and low spending are budget customers
- Customers with high income but low spending are potential customers
- Customers with low income but high spending show unique behavior
- Each cluster represents a different type of customer group

7. Visualization

A scatter plot was created to visualize customer segments.

Each cluster is represented using different colours, making it easy to understand customer distribution.

8. Conclusion

Customer segmentation helps businesses:

- Understand customer behaviour
- Identify valuable customers
- Create targeted marketing strategies

9. Future Scope

- Use more features like Age and Gender
- Apply advanced clustering techniques
- Build interactive dashboards for better analysis